GATE CS — Digital Logic

Topic-wise past questions (2016–2026) + new practice questions with solutions

Built from the papers in pAPER/ of the GATE2027 repo. Topics are ordered by total marks (most important first).

Topic	Past questions	Total marks	New practice Qs
Boolean algebra & K-map	23	37	2
Number representation, IEEE-754, arithmetic	27	35	2
Sequential circuits (flip-flops, counters, FSM)	11	18	2
Combinational circuits (MUX/decoder)	8	13	2
Total	**69**	**103**	**8**

How to read this

- Each topic has **A. Past-paper questions** (newest first; tagged year · set · question number · marks) and **B. New practice questions** with short solutions.
- Past questions have **no answer key** here; check them against the official answer keys.
- Question text was extracted from the PDFs. Code, tables, symbols and figures may be flattened or lost. Entries marked [OCR] come from scanned papers and may contain errors; open the original in pAPER/<year>/ for the exact wording.
- New practice questions are original, in GATE style; numerical and program-output answers were verified.

Boolean algebra & K-map

K-map and minterm minimization, canonical forms, identities, gate counts. · 23 questions, 37 marks over 15 papers (≈2.5 marks/paper)

A. Past-paper questions (23)

2026 CS1 · Q.21 · 1 mark

Consider the following Boolean expression of a function F : F(P, Q) = ($\overline{P}$ + Q) ⊕ ($\overline{P}$Q) Which of the following expressions is/are equivalent to F ?

(A) P⊕Q

(B) P⊕Q

(C) $\overline{P}$ ⊕ Q

(D) $\overline{P}$ ⊕ $\overline{Q}$

2026 CS1 · Q.48 · 2 marks

Consider a Boolean function F with the following minterm expression: F(P, Q, R, S) = ∑ m (1, 2, 3, 4, 5, 7, 10, 12, 13, 14) Which of the following options is/are the minimal sum-of-products expression(s) of F ?

(A) $\overline{P}$S + Q$\overline{R}$ + $\overline{P}$$\overline{Q}$ R + $\overline{Q}$ R$\overline{S}$

(B) $\overline{P}$S + Q$\overline{R}$ + $\overline{P}$$\overline{Q}$ R + PR$\overline{S}$

(C) $\overline{P}$S + Q$\overline{R}$ + PQ$\overline{S}$ + PR$\overline{S}$

(D) $\overline{P}$S + Q$\overline{R}$ + PQ$\overline{S}$ + $\overline{Q}$ R$\overline{S}$

2026 CS2 · Q.16 · 1 mark

Which one of the following options is not a property of Boolean Algebra? Note: + is OR operation, . is AND operation, and ′ is NOT operation

(A) a+b =b+a

(B) a . a′ = 1

(C) a + a′ = 1

(D) a .b = b .a

2026 CS2 · Q.40 · 2 marks

Consider the following 4-variable Boolean function F(A, B, C, D) = Σm(0, 1, 2, 3, 8, 9, 10, 11) Consider A as MSB, D as LSB. Which one of the following options represents the minimal sum of products form for the above function? Note: + is OR operation, . is AND operation, ′ is NOT operation

(A) A′ + B′ + C ′ + D′

(B) B′

(C) A′ . B′ + A . B

(D) A′

2025 CS1 · Q.24 · 1 mark

Let X be a 3-variable Boolean function that produces output as ‘1’ when at least two of the input variables are ‘1’. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?

(A) X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)

(B) X(a, b, X(a, b, c)) = X(a, b, c)

(C) X(a, b, X(a, c, d)) = (X(a, b, a) AND X(c, d, c))

(D) X(a, b, c) = X(a, X(a, b, c), X(a, c, c))

2025 CS1 · Q.42 · 2 marks

Consider the following four variable Boolean function in sum-of-product form F(b3 , b2 , b1 , b0) = ∑(0, 2, 4, 8, 10, 11, 12). where the value of the function is computed by considering b3 b2 b1 b0 as a 4-bit binary number, where b3 denotes the most significant bit and b0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F?

(A) $\bar{b}1\ \bar{b}0 + \bar{b}2\ \bar{b}0 + b1\ \bar{b}2\ b3$

(B) $\bar{b}1\ \bar{b}0 + \bar{b}2\ \bar{b}0$

(C) $\bar{b}2\ \bar{b}0 + b1\ b2\ b3$

(D) $\bar{b}0\ \bar{b}2 + \bar{b}3$

2025 CS2 · Q.31 · 1 mark

Consider the following logic circuit diagram. Which is/are the CORRECT option(s) for the output function F?

(A) $\overline{XY}$

(B) $\bar{X} + \bar{Y} + X\bar{Y}$

(C) $\overline{X\,Y} + \bar{X} + X\bar{Y}$

(D) $X + \bar{Y}$

2025 CS2 · Q.43 · 2 marks

Given the following Karnaugh Map for a Boolean function F(w, x, y, z): Which one or more of the following Boolean expression(s) represent(s) F?

(A) $w\bar{x}\ \bar{y}\bar{z} + w\bar{x}\ \bar{y}\bar{z} + w\bar{x}\ y\bar{z} + w\bar{x}\ y\bar{z} + xz$

(B) $w\bar{x}\ \bar{y}\bar{z} + w\bar{x}\ y\bar{z} + w\bar{x}\ yz + xz$

(C) $w\bar{x}\ \bar{y}\bar{z} + w\bar{x}\ \bar{y}\bar{z} + w\bar{x}\ \bar{y}z + xz$

(D) $\bar{x}\ \bar{z} + xz$

2025 CS2 · Q.50 · 2 marks

Which of the following Boolean algebraic equation(s) is/are CORRECT?

(A) $\bar{A}\ B\ C + A\ \bar{B}\ \bar{C} + \bar{A}\ \bar{B}\ \bar{C} + A\ \bar{B}\ C + A\ B\ C = B\ C + \bar{B}\ \bar{C} + \bar{A}\ \bar{B}$

(B) $A\ B + \bar{A}\ C + B\ C = A\ B + \bar{A}\ C$

(C) $(A + C)\ (\bar{A} + B) = A\ B + \bar{A}\ C$

(D) $\overline{(A + \bar{B} + D)}\ (C + D)\ (\bar{A} + C + D)\ \overline{(A + B + D)} = \bar{A}\ D + \bar{C}\ \bar{D}$

2024 CS1 · Q.47 · 2 marks

Consider a Boolean expression given by F(X, Y, Z) = ∑(3,5,6,7). Which of the following statements is/are CORRECT?

(A) F(X, Y, Z) = Π(0,1,2,4)

(B) F(X, Y, Z) = XY + YZ + XZ

(C) F(X, Y, Z) is independent of input Y

(D) F(X, Y, Z) is independent of input X

2024 CS2 · Q.30 · 1 mark

For a Boolean variable x, which of the following statements is/are FALSE?

(A) x. 1 = x

(B) x+1=x

(C) x. x = 0

(D) $x + \bar{x} = 1$

2024 CS2 · Q.50 · 2 marks
Consider 4-variable functions f1, f2, f3, f4 expressed in sum-of-minterms form as given below. f1 = ∑(0,2,3,5,7,8,11,13) f2 = ∑(1,3,5,7,11,13, 15) f3 = ∑(0,1,4,11) f4 = ∑(0,2,6,13) With respect to the circuit given above, which of the following options is/are CORRECT?
(A) Y = ∑(0,1,2,11,13)
(B) Y = Π(3,4, 5,6,7,8,9,10,12,14,15)
(C) Y = ∑(0,1,2,3,4,5,6,7)
(D) Y = Π(8,9,10,11,12,13,14,15)

2021 · Q.42 · 2 marks [OCR]
Consider the following Boolean expression. F=(X+Y+Z)(X+Y)(¥+Z) Which of the following Boolean expressions is/are equivalent to F (complement of F)?
(A) | (X+YV4+Z)(X+Y)(¥+Z
(B) | XY +Z
(C) | (X+Z)(Y +7)
(D) | XY+YZ+XY

2020 · Q.28 · 2 marks [OCR]
Consider the Boolean function z(a, b,c). a = Which one of the following minterm lists represents the circuit given above? W 7s Yo 1,3,7) ® 7s Ya 4,5, 6,7) O gs Ye 4,5,6,7) O gs Ye 3,5)

2019 · Q.6 · 1 mark [OCR]
Which one of the following is NOT a valid identity?
(A) (x ® y) ®z = xO (y Gz)
(B) k+y) ®z = x ®(y+z) (C)x@y = x+y, ifxy=0 (D)x®y = (@ytx'y'y' Q7_ IfL is a regular language over X = {a,b}, which one of the following languages is NOT regular ?
(A) L- LR = {xy |x € L,y® € L}
(B) {ww | w EL}
(C) Prefix (L) = {x € Z*|Ay € &* such that xy € L}
(D) Suffix (L) = {y € 2*|Ax € &* such that xy € L}

2019 · Q.30 · 2 marks [OCR]
Consider three 4-variable functions fi, f2, and f3, which are expressed in sum-of-minterms as fi == (0, 2, 5, 8, 14), h= 2 (2,3,6,8,14,15), f= (2,7, 11, 14) For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:
(A) =(7,8, 11)
(B) ©2,7,8, 11, 14)
(C) X22, 14)
(D) = (0, 2, 3, 5, 6, 7, 8, 11, 14, 15)

2019 · Q.50 · 2 marks [OCR]
What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as f = > (0, 2, 5, 7, 8, 10, 13, 15)? Assume that all the inputs and their complements are available. Answer:

2018 · Q.4 · 1 mark
Let ⊕ and ⊙ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?
(A) P⊕Q =P⊙Q
(B) $\bar{P} \oplus Q = P \odot Q$
(C) $\bar{P} \oplus \bar{Q} = P \oplus Q$
(D) $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

2018 · Q.49 · 2 marks
Consider the minterm list form of a Boolean function F given below. F(P, Q, R, S) = ∑ m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14) Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____. CS 17/20 GATE 2018

2017 CS1 · Q.21 · 1 mark [OCR]
Consider the Karnaugh map given below. where X represents "don't care" and blank represents 0. — 00 O01 11 10 00 x x 01 1 x ll 1 1 10 x De Assume for all inputs (a, 5, c, d), the respective complements (ab td) are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is

2017 CS2 · Q.27 · 2 marks [OCR]
If w, x,y,z are Boolean variables. then which one of the following is INCORRECT?
(A) wx+w(xty)t+x(x+y) =x+wy
(B) wX(y+ Z)+Wx=W+x+yz
(C) (Wk(y + xZ) + WX)y = x¥
(D) (w + y)(wxy + wyz) = wxy + wyz =====PAGE 11=====

2017 CS2 · Q.28 · 2 marks [OCR]
Given f(w,x,y,Z) = Ym(0,1,2,3,7,8,10) + Y4(5,6,11,15), where d represents the don't-care condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of f(w, x,y,z)?
(A) f = (@ + IE +2)
(B) f = (W+2)(x +z)
(C) f = (w4+z)(X¥ +2)
(D) f = (w+2Z)(¥ +z)

2016 · Q.6 · 1 mark
Consider the Boolean operator # with the following properties: x # 0 = x, x # 1 = $\bar{x}$, x # x = 0 and x # $\bar{x}$ = 1. Then x # y is equivalent to
(A) $x\bar{y} + \bar{x}y$
(B) $x\bar{y} + \bar{x}\bar{y}$
(C) $\bar{x}y + xy$
(D) $xy + \bar{x}\bar{y}$

B. New practice questions with solutions

1. (1 mark · NAT) The minimum number of literals in a minimal sum-of-products expression for F(A, B, C) = Σm(0, 1, 2, 5, 6, 7) is _____.

6. One minimal cover is A′B′ + BC′ + AC, which has 3 terms of 2 literals each. The K-map forms

a cycle of six overlapping pairs, and no cover needs fewer than 6 literals.

2. (2 marks · MCQ) For variables (w, x, y, z), F = Σm(0, 2, 5, 7, 8, 10, 13, 15) equals

A. x ⊕ z (B) x ⊙ z (C) w ⊕ y (D) w ⊙ y

(B). The minterms are exactly the cases where x = z (x′z′ covers 0, 2, 8, 10 and xz covers 5, 7, 13, 15). So F is the XNOR of x and z.

Number representation, IEEE-754, arithmetic

2's complement/sign-magnitude, overflow, IEEE-754, radix conversion, adders. · 27 questions, 35 marks over 15 papers (≈2.3 marks/paper)

A. Past-paper questions (27)

2026 CS1 · Q.22 · 1 mark
Consider the 8-bit signed integers X, Y and Z represented using the sign-magnitude form. The binary representations of X and Y are as follows: X: 10110100 Y: 01001100 Which of the following operations to compute Z result(s) in an arithmetic overflow?
(A) Z =X+Y
(B) Z =X−Y
(C) Z = −X + Y
(D) Z = −X − Y

2026 CS1 · Q.36 · 2 marks
Consider the real valued variables X, Y and Z represented using the IEEE 754 single-precision floating-point format. The binary representations of X and Y in hexadecimal notation are as follows: X: 35C00000 Y: 34A00000 Let Z = X + Y. Which one of the following is the binary representation of Z, in hexadecimal notation?
(A) 35C80000
(B) 35CC0000
(C) 35E80000
(D) 35EC0000

2026 CS2 · Q.28 · 1 mark
In a system, numbers are represented using 4-bit two's complement form. Consider four numbers N1 =1011, N2 =1101, N3 =1010 and N4 =1001 in the system. Which of the following operations will result in arithmetic overflow?
(A) N1 + N2
(B) N2 + N3
(C) N3 − N4
(D) N1 + N4

2026 CS2 · Q.34 · 1 mark
The 32-bit IEEE 754 single precision representation of a number is 0xC2710000. The number in decimal representation is _____. (rounded off to two decimal places)

2025 CS1 · Q.25 · 1 mark
The number −6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of −6?
(A) 1000 1010 in 8-bits
(B) 1111 1010 in 8-bits
(C) 1000 0000 0000 1010 in 16-bits
(D) 1111 1111 1111 1010 in 16-bits

2025 CS2 · Q.32 · 1 mark
The following two signed 2’s complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth’s algorithm: M: 1100 1101 1110 1101 and Q: 1010 0100 1010 1010 The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

2025 CS2 · Q.49 · 2 marks
Three floating point numbers X, Y, and Z are stored in three registers RX, RY, and RZ, respectively in IEEE 754 single precision format as given below in hexadecimal: RX = 0xC1100000, RY = 0x40C00000, and RZ = 0x41400000 Which of the following option(s) is/are CORRECT?
(A) 4(X + Y) + Z = 0
(B) 2Y – Z = 0
(C) 4X + 3Z = 0
(D) X+Y+Z = 0

2024 CS1 · Q.13 · 1 mark
Consider a system that uses 5 bits for representing signed integers in 2’s complement format. In this system, two integers A and B are represented as A=01010 and B=11010. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?
(A) A+B
(B) A−B
(C) B−A
(D) 2∗B

2024 CS2 · Q.14 · 1 mark
The format of a single-precision floating-point number as per the IEEE 754 standard is: Sign Exponent Mantissa (1bit) (8 bits) (23 bits) Choose the largest floating-point number among the following options. Sign Exponent Mantissa 0 0111 1111 1111 1111 1111 1111 1111 111
(A) Sign Exponent Mantissa 0 1111 1110 1111 1111 1111 1111 1111 111
(B) Sign Exponent Mantissa 0 1111 1111 1111 1111 1111 1111 1111 111
(C) Sign Exponent Mantissa
(D) 0 0111 1111 0000 0000 0000 0000 0000 000

2024 CS2 · Q.49 · 2 marks
Which of the following is/are EQUAL to 224 in radix-5 (i.e., base-5) notation?
(A) 64 in radix-10
(B) 100 in radix-8
(C) 50 in radix-16
(D) 121 in radix-7

2023 · Q.32 · 1 mark
A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is . CS

2023 · Q.45 · 2 marks
Consider the IEEE-754 single precision floating point numbers P=0xC1800000 and Q=0x3F5C2EF4. Which one of the following corresponds to the product of these numbers (i.e., P × Q), represented in the IEEE-754 single precision format?
(A) 0x404C2EF4
(B) 0x405C2EF4
(C) 0xC15C2EF4
(D) 0xC14C2EF4 CS

2022 · Q.18 · 1 mark
Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation R1+R2, which one of the following values of R1 and R2 gives an arithmetic overflow?
(A) R1 = 1011 and R2 = 1110
(B) R1 = 1100 and R2 = 1010
(C) R1 = 0011 and R2 = 0100
(D) R1 = 1001 and R2 = 1111

2022 · Q.41 · 2 marks
Consider three floating point numbers A , B and C stored in registers RA, RB and RC, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows. RA= 0xC1400000 RB = 0x42100000 RC = 0x41400000 Which one of the following is FALSE?
(A) AC 0
(B) C A B
(C) B 3C
(D) (B C) 0

2021 · Q.6 · 1 mark [OCR]
Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number?
(A) | 15
(B) | 21 (c) | D2
(D) | 528 CS, Set-1 - Copyright © GATE 2021

2021 · Q.24 · 1 mark [OCR]
Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127. fo E : 10000001 F : 11110000000000000000000 Here S, E and F denote the sign, exponent and fraction components of the floating point representation. The decimal value corresponding to the above representation (rounded to 2 decimal places) is

2020 · Q.29 · 2 marks [OCR]
Consider three registers R1, R2, and R3 that store numbers in IEEE-754 single precision floating point format. Assume that R1 and R2 contain the values (in hexadecimal notation) 0x42200000 and 0xC1200000, respectively. If R3= -. what is the value stored in R3?
(A) 0x40800000 (8) 0xC0800000 © 0x83400000 (0) 0xC8500000

2019 · Q.4 · 1 mark [OCR]
In 16-bit 2’s complement representation, the decimal number —28 is:
(A) 1111 1111 0001 1100
(B) 0000 0000 1110 0100
(C) 1111 1111 1110 0100
(D) 1000 0000 1110 0100

2019 · Q.8 · 1 mark [OCR]
Consider Z = X — Y, where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:
(A) n bits
(B) n— 1 bits
(C) n+ 1 bits
(D) n+ 2 bits

2018 · Q.33 · 2 marks
Consider the unsigned 8-bit fixed point binary number representation below, b7 b6 b5 b4 b3 . b2 b1 b0 where the position of the binary point is between b3 and b2. Assume b7 is the most significant bit. Some of the decimal numbers listed below cannot be represented exactly in the above representation: (i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001 Which one of the following statements is true?
(A) None of (i), (ii), (iii), (iv) can be exactly represented
(B) Only (ii) cannot be exactly represented
(C) Only (iii) and (iv) cannot be exactly represented
(D) Only (i) and (ii) cannot be exactly represented CS 10/20 GATE 2018

2017 CS1 · Q.7 · 1 mark [OCR]
The n-bit fixed-point representation of an unsigned real number Yuses /bits for the fraction part. Let i=n— f. The range of decimal values for Xin this representation is
(A) 2°F to 2!
(B) 27 to(2'—2-7)
(C) 0to2!
(D) Oto (2! — 2-7) =====PAGE 5=====

2017 CS1 · Q.9 · 1 mark [OCR]
When two 8-bit numbers 4,---.4, and B,---B, in 2’s complement representation (with 4, and B, as the least significant bits) are added using a ripple-carry adder. the sum bits obtained are S,-++S, and the carry bits are C,---C, . An overflow is said to have occurred if
(A) the carry bit C; is 1
(B) all the carry bits (C,.---.C,) are 1
(C) (4,:B,+S,+4,-B,°S,)is1
(D) (4):By So +p ‘By: So) is 1 =====PAGE 6=====

2017 CS2 · Q.1 · 1 mark [OCR]
The representation of the value of a 16-bit unsigned integer Y in hexadecimal number system is BCA9. The representation of the value of Yin octal number system is
(A) 571244
(B) 736251
(C) 571247
(D) 136251

2017 CS2 · Q.12 · 1 mark [OCR]
Given the following binary number in 32-bit (single precision) IEEE-754 format: 00111110011011010000000000000000 The decimal value closest to this floating-point number is
(A) 1.45 x 10+
(B) 1.45 x 107+ (C):227' X10 4
(D) 2.27 x 10+

2017 CS2 · Q.24 · 1 mark [OCR]
Consider a quadratic equation x? — 13x + 36 = 0 with coefficients in a base b. The solutions of this equation in the same base b are x = 5 and x = 6. Then b=

2016 · Q.7 · 1 mark
The 16-bit 2’s complement representation of an integer is 1111 1111 1111 0101; its decimal representation is .

2016 · Q.33 · 2 marks
Consider a carry lookahead adder for adding two n-bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is
(A) Θ(1)
(B) Θ(log(n)) √
(C) Θ(n)
(D) Θ(n)

B. New practice questions with solutions

1. (1 mark · NAT) The decimal value of the 8-bit two’s complement number 10110110 is _____.

−74. Inverting gives 01001001, adding 1 gives 01001010 = 74, so the value is −74.

2. (2 marks · NAT) The 32-bit IEEE-754 single precision number 0xC1A40000 has the decimal value _____.

−20.5. The sign bit is 1. The exponent field is 10000011 = 131, so the exponent is 4. The mantissa is .0100100…, so the value is $-1.28125 \times 2^4 = -20.5$.

Sequential circuits (flip-flops, counters, FSM)

Counters and flip-flop circuits, state sequences, FSM tables. · 11 questions, 18 marks over 15 papers (≈1.2 marks/paper)

A. Past-paper questions (11)

2026 CS1 · Q.37 · 2 marks

Consider a 2-bit saturating up/down counter that performs the saturating up count when the input P is 0, and the saturating down count when P is 1. The Next State table of the counter is as shown. The counter is built as a synchronous sequential circuit using D flip-flops. Input Current Next State State P Q1 Q0 Q1+ Q0+ 0 0 0 0 1 0 0 1 1 0 0 1 0 1 1 0 1 1 1 1 1 0 0 0 0 1 0 1 0 0 1 1 0 0 1 1 1 1 1 0 Which one of the following options corresponds to the expressions for the inputs of the D flip-flops, D1 and D0 ?

(A) D1 = P Q1 + $\bar{P}$Q0 + Q1 Q0 D0 = P Q0 + $\bar{P}$ Q1 + Q1$\overline{\equiv}$ Q0

(B) D1 = $\bar{P}$ Q1 + $\bar{P}$Q0 + Q1 Q0 D0 = $\bar{P}\overline{\equiv}$ Q0 + $\bar{P}$ Q1 + Q1$\overline{\equiv}$ Q0

(C)$\overline{\equiv}$1 + $\bar{P}$ Q0 + Q1 Q0 D1 = $\bar{P}$ Q D0 = $\bar{P}$ Q0 + $\bar{P}$ Q1 + Q1$\overline{\equiv}$ Q0

(D)$\overline{\equiv}$1 + $\bar{P}$ Q0 + Q1 Q0 D1 = P Q D0 = P$\overline{\equiv}$ Q0 + $\bar{P}$ Q1 + Q1$\overline{\equiv}$ Q0

2025 CS1 · Q.59 · 2 marks

Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer) Present state Next state Output f X=0 X=1 X=0 X=1 A F B 0 0 B D C 0 0 C F E 0 0 D G A 1 0 E D C 0 0 F F B 1 1 G G H 0 1 H G A 1 0

2025 CS1 · Q.60 · 2 marks

Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____ . (Answer in integer)

2025 CS2 · Q.34 · 1 mark

In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)

2023 · Q.21 · 1 mark

The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure. Multiplexer 0 Q 1 S Match the functional equivalence of this circuit to one of the following options.

(A) D Flip-flop

(B) D Latch

(C) Half-adder

(D) Demultiplexer CS

2023 · Q.43 · 2 marks

Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning,

2021 · Q.28 · 2 marks [OCR]
Consider a 3-bit counter, designed using T flip-flops, as shown below: Lr so Ta aL Ta a4 Clock Pulse P’ Q rR Assuming the initial state of the counter given by PQR as 000, what are the next three states?
(A) | 011, 101, 000
(B) | 001, 010, 111
(C) | 011, 101, 111
(D) | 001, 010, 000

2018 · Q.22 · 1 mark
Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops. in out D Q D Q clock The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of “in” is _____. CS 6/20 GATE 2018

2017 CS1 · Q.33 · 2 marks [OCR]
Consider a combination of T and D flip-flops connected as shown below. The output of the D flip- flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop. Clock Initially. both Qo and Qi are set to 1 (before the 1* clock cycle). The outputs
(A) Q1:Qo after the 3 cycle are 11 and after the 4" cycle are 00 respectively
(B) Q:Qo after the 3 cycle are 11 and after the 4" cycle are 01 respectively
(C) Q1Qo after the 3° cycle are 00 and after the 4" cycle are 11 respectively
(D) Q:Qo after the 3 cycle are 01 and after the 4" cycle are 01 respectively

2017 CS2 · Q.42 · 2 marks [OCR]
The next state table of a 2-bit saturating up-counter is given below. Q. %| QF 0 0/0 1 Oo) | 0 L © |1 1 | 1 The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T, and Ty are (A)T, = Q1 9, Ty = Q1Q
(B) T, = Q1Q, Ty = Q1 + Q (C)T, = Q14+ Qo, To = Q1+ Qo
(D) T; = Q1Qo, Ty = Q1 + Qo =====PAGE 17=====

2016 · Q.8 · 1 mark
We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is .

B. New practice questions with solutions

1. (1 mark · NAT) A 4-bit binary ripple counter is driven by a 64 kHz clock. The frequency of the most significant output in kHz is _____.

4. Each stage halves the frequency, so the MSB output is 64 / 2^4 = 4 kHz.

2. (2 marks · MCQ) A circuit has two D flip-flops with $D_1 = Q_1 \oplus Q_0$ and $D_0 = Q_1'$. It starts in state $Q_1Q_0 = 00$. After 100 clock pulses the state Q_1Q_0 is

A. 00 (B) 01 (C) 10 (D) 11

(B). The sequence is 00 → 01 → 11 → 00, a cycle of length 3. 100 mod 3 = 1, so the state is 01.

Combinational circuits (MUX/decoder)

MUX/decoder/adder-based circuit analysis. · 8 questions, 13 marks over 15 papers (≈0.9 marks/paper)

A. Past-paper questions (8)

2026 CS2 · Q.59 · 2 marks

Consider the digital circuit shown below with two input lines A and B, two select lines S0 and S1, and an output line Y. The blocks Q and M represent active high 2:4 decoder and 4-to-1 multiplexer, respectively. Out of 16 possible input combinations, the number of combinations that produce Y=1 is _____. (answer in integer) Note: One input combination is an instance of [A B S1 S0].

2024 CS1 · Q.28 · 1 mark

Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?
(A) With no propagation delays, the output Y is always logic Zero
(B) With no propagation delays, the output Y is always logic One With propagation delays, the output Y can have a transient logic One after X
(C) transitions from logic Zero to logic One With propagation delays, the output Y can have a transient logic Zero after X
(D) transitions from logic One to logic Zero

2024 CS1 · Q.64 · 2 marks

Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively. X1 0

2023 · Q.44 · 2 marks

A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0–X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown. Multiplexer X0 0 M1 X1 1 Q X2 2 Multiplexer X3 3 S1 S0 0 A M3 C Q Multiplexer X4 0 1 S M2 X5 1 Q X6 2 X7 3 S1 S0 A C B Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function A + A.C + A.B.C?
(A) (1, 1, 0, 0, 1, 1, 1, 0)
(B) (1, 1, 0, 0, 1, 1, 0, 1)
(C) (1, 1, 0, 1, 1, 1, 0, 0)
(D) (0, 0, 1, 1, 0, 1, 1, 1) CS

2022 · Q.40 · 2 marks
Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T? Code Word M S-address R-address P Q S R T X DDS
(A) P is 10:1 multiplexer; Q is 5:1 multiplexer; T is 2:1 multiplexer
(B) P is 10:210 decoder; Q is 5:25 decoder; T is 2:1 encoder
(C) P is 10:210 decoder; Q is 5:25 decoder; T is 2:1 multiplexer
(D) P is 1:10 de-multiplexer; Q is 1:5 de-multiplexer; T is 2:1 multiplexer

2020 · Q.19 · 1 mark [OCR]
A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The minimum number of select lines needed for the multiplexer is

2020 · Q.20 · 1 mark [OCR]
If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of m + n is j

2016 · Q.30 · 2 marks
Consider the two cascaded 2-to-1 multiplexers as shown in the figure. R 0 0 2−to−1 0 2−to−1 X R 1 MUX 1 MUX s s P Q The minimal sum of products form of the output X is
(A) $\bar{P}\,\bar{Q} + PQR$
(B) $\bar{P}Q + QR$
(C) $PQ + \bar{P}\,\bar{Q}R$
(D) $\bar{Q}\,\bar{R} + PQR$

B. New practice questions with solutions

1. (1 mark · MCQ) A 4:1 multiplexer has select lines S_1S_0 and data inputs $I_0 = 0$, $I_1 = 1$, $I_2 = 1$, $I_3 = 0$. The output is

A. $S_1 \cdot S_0$ (B) $S_1 + S_0$ (C) $S_1 \oplus S_0$ (D) $S_1 \odot S_0$

(C). The output is 1 for 01 and 10 only, which is XOR.

2. (2 marks · MCQ) F(A, B, C) = Σm(1, 2, 4, 7) is implemented with a 4:1 MUX with A and B as select lines (A is the MSB). The inputs I_0, I_1, I_2, I_3 are, in order,

A. C, C′, C′, C
B. C′, C, C, C′
C. C, C, C′, C′
D. 0, 1, 1, 0

(A). For AB = 00 the minterm 1 gives C. For AB = 01 the minterm 2 gives C′. For AB = 10 the minterm 4 gives C′. For AB = 11 the minterm 7 gives C.

Tip: attempt each question on paper, check the answer, and then rework any you missed one to two days later. Use `GATE_CS_Topicwise_Questions.md` for the real past-paper questions on the same topics.